Relational schema

`K = (pid, cid, sectionnumber, term)` is the section key. SQL names are lowercase.

Tables and keys

Relation	Primary key	Foreign keys
`professor(pid, name)`	`pid`	None
`course(cid, coursenumber, unit)`	`cid`	None
`department(did, name, manageby)`	`did`	`(manageby, did) -> worksin(pid, did)`
`worksin(pid, did)`	`(pid, did)`	`pid -> professor(pid)`; `did -> department(did)`
`student(sid, name, dept, unitrequirement)`	`sid`	`dept -> department(did)`
`teach(K, dept)`	`K`	`cid -> course(cid)`; `(pid, dept) -> worksin(pid, did)`
`assignment(K, id, weight, duedate)`	`(K, id)`	`K -> teach(K)`
`enroll(K, sid)`	`(K, sid)`	`K -> teach(K)`; `sid -> student(sid)`
`assign(K, id, sid, grade, turnindate)`	`(K, id, sid)`	`(K, id) -> assignment(K, id)`; `(K, sid) -> enroll(K, sid)`

Department `name` is the only additional candidate key. Other names and course numbers can repeat.

Result foreign keys must match the assignment and enrollment in the same section. Managers/professors need matching department membership. The manager FK is `DEFERRABLE INITIALLY DEFERRED` to allow both rows to be inserted before commit. Deletes do not cascade.

Types and constraints

- IDs and section numbers: positive integers.
- Names and terms: nonblank text; department names are unique.
- Course units: integer 1-12. Individual unit requirements: integer 1-300.
- Weight: `NUMERIC(5,2)`, greater than 0 and at most 100.
- Grade: `NUMERIC(5,2)`, 0-100 when present.
- Dates: `DATE`. Only `duedate`, `grade`, and `turnindate` allow NULL.
- Primary keys are unique and NOT NULL. Foreign keys require matching parents.

Functional dependencies

Relation	Dependencies
professor	pid -> name
course	cid -> coursenumber, unit
department	did -> name, manageby; name -> did, manageby
student	sid -> name, dept, unitrequirement
teach	K -> dept
assignment	(K, id) -> weight, duedate
assign	(K, id, sid) -> grade, turnindate
worksin, enroll	All columns are in the primary key; no non-key attributes.

dept does not determine unitrequirement. A section number alone is not a key.

Normalization

The tables are in 3NF: non-key attributes depend on the whole key, with no non-key-to-non-key dependencies. Every listed determinant is a candidate key, so these dependencies also meet BCNF.

Names, units, and assignment details are stored once, avoiding repeated updates across grades. Sections can be inserted before enrollments and assignments before results. Deleting a result does not delete its assignment.

Final grades/GPA are calculated to avoid stale totals. The enrollment_grades view calculates weights, earned points, current grades, and final grades.

enroll_sid_idx supports a student's course history. worksin_did_idx supports staff lookups by department. Primary keys already index section/assignment keys.

Rules handled by the application

- A CHECK cannot enforce weights totaling 100 across rows. Phase 2 will lock teach during assignment edits/finalization and check the total. The view returns finals only for fully graded, 100-weight schemes.
- What-if calculations must not overwrite recorded grades.
- Phase 2 handles access to individual grades and aggregate peer reports.
- Grade/GPA formulas and repeated-attempt rules are in Requirements document.

Changes from the draft

Use sid for students, add id to assign, and include term in section/dependent keys. dept is non-key. Grades are decimal percentages; unit requirements are per student.